

ASD Longitudinal RMD

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This Markdown summarizes information regarding the data that will be used for a longitudinal analysis of vocabulary acquisition in autistic children

Demographic Information

Demographics for kids with between 3 & 600 productive vocabulary

```
## [1] "Number of unique subjectkeys is 349 and the number of unique CDI entries is 628"
```

```
##  subjectkey      interview_age      nProduced
##  Length:409436   Min.    : 8.00   Min.    : 3.0
##  Class :character 1st Qu.: 26.00   1st Qu.: 31.0
##  Mode  :character Median : 36.00   Median :151.0
##                      Mean   : 43.23   Mean   :184.6
##                      3rd Qu.: 54.00   3rd Qu.:296.0
##                      Max.    :173.00   Max.    :582.0
```

Longitudinal Analysis

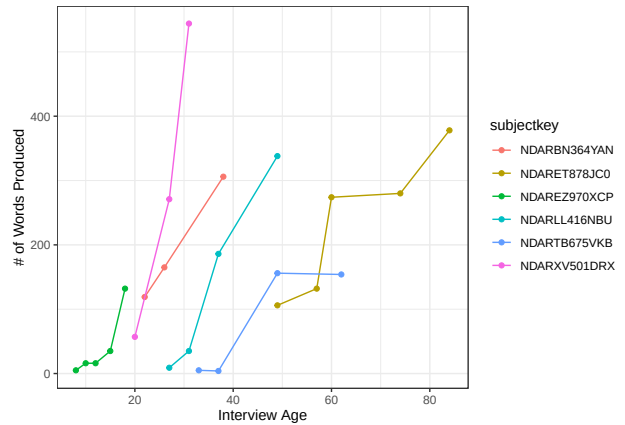
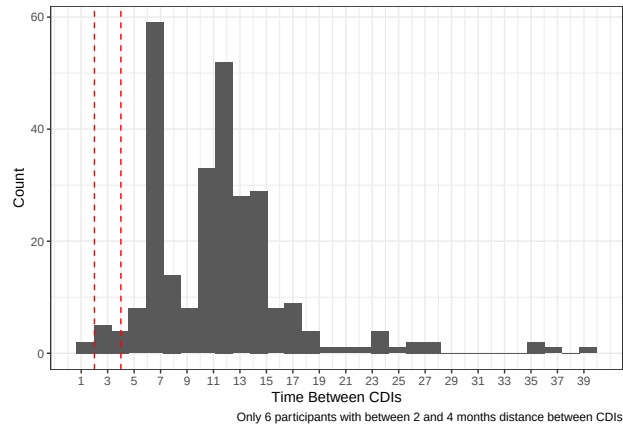
Of the 628 CDIs, 206 only have an entry for a single age, removing them from longitudinal consideration. That leaves 422(111 unique kids) with at least 2 CDIs.

Table 1: CDI Longitudinal Counts

# of CDIs	# of Records	# of Kids
1	206	206
2	102	51
3	162	54
4	128	32
5	30	6

Kids with longitudinal data in 3 month intervals

subjectkey	interview_age	diff	nProduced
NDARBN364YAN	22	NA	119
NDARBN364YAN	26	4	165
NDARBN364YAN	38	12	306
NDARET878JC0	49	NA	106
NDARET878JC0	57	8	132
NDARET878JC0	60	3	274
NDARET878JC0	74	14	280
NDARET878JC0	84	10	378
NDAREZ970XCP	8	NA	5
NDAREZ970XCP	10	2	16
NDAREZ970XCP	12	2	16
NDAREZ970XCP	15	3	35
NDAREZ970XCP	18	3	132
NDARLL416NBU	27	NA	9
NDARLL416NBU	31	4	35
NDARLL416NBU	37	6	186
NDARLL416NBU	49	12	338
NDARTB675VKB	33	NA	5
NDARTB675VKB	37	4	4
NDARTB675VKB	49	12	156
NDARTB675VKB	62	13	154
NDARXV501DRX	20	NA	57
NDARXV501DRX	27	7	271
NDARXV501DRX	31	4	544



Kids with longitudinal data in 12 month intervals

There are 111 kids with CDIs at the specificity of 9 - 15 months between CDIs. Below are tables and figures showing the distribution of time between CDIs both across participants and how many individual participants have multiple CDIs within the 9 - 15 time range.

```
## [1] "There are 111 unique kids with CDIs 9 - 15 months between CDIs"
```

```
## [1] "There are 150 cases of CDIs being between 9 and 15 months apart. The counts are shown below."
```

Table 3: CDI Latencies Counts

Month Latency	Count
9	8
10	14
11	19
12	52
13	28
14	15
15	14

Table 4: Multiple Contributions Counts

# of Contributions	Count
1	79
2	25
3	7

Histograms

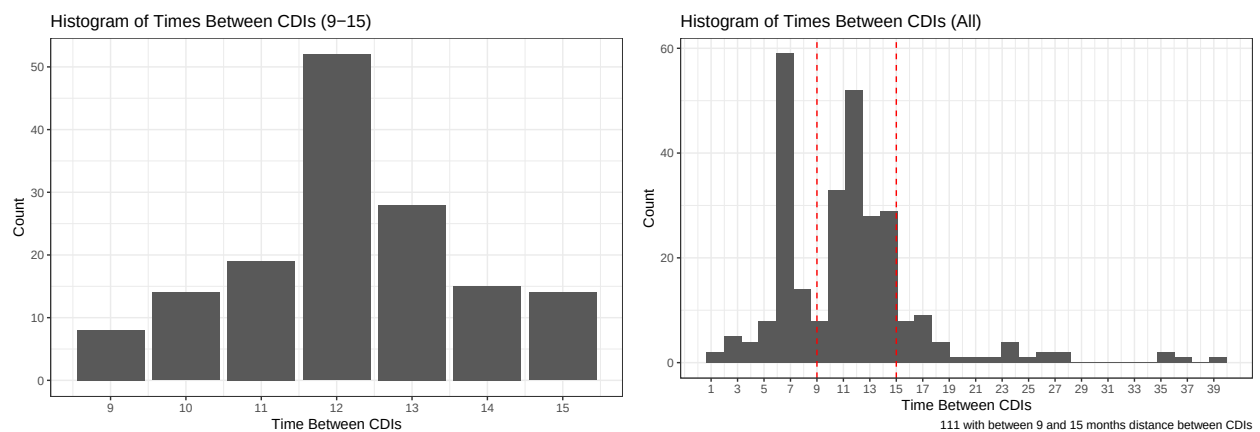


Figure 1: Histograms of 9 - 15 months between CDIs

Spaghetti Plots

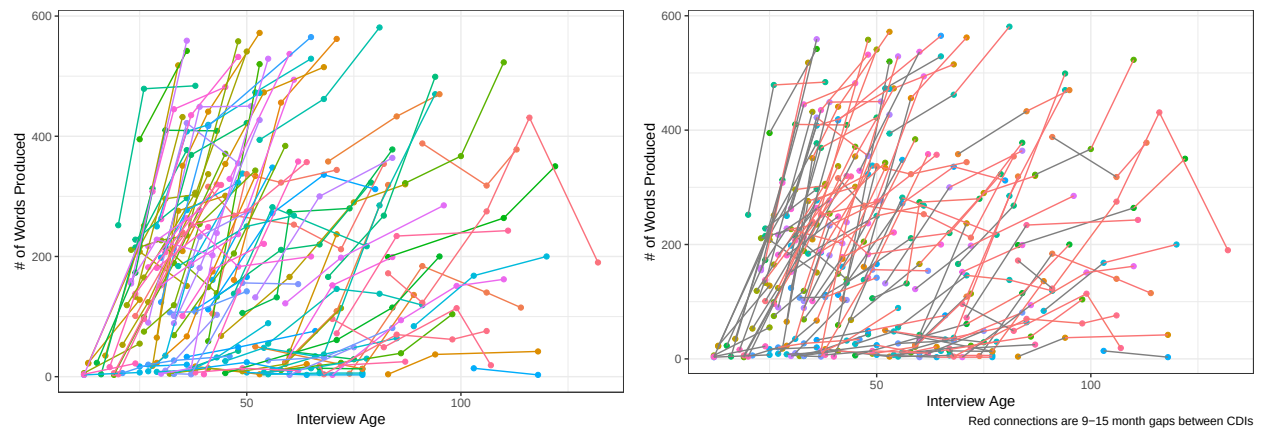


Figure 2: Spaghetti plots of longitudinal vocabulary growth